

ROOT

The diagram illustrates the hardware and timing for a program counter system. It includes a logic schematic and a timing diagram.

Logic Schematic:

- TEST_PROGRAM_COUNTER:** A block with an ON/OFF input and DBUS[0..7] data bus. It has internal signals CLK, CE, IE, DOE, AOE, and RESET.
- PROGRAM_COUNTER (PC):** A block with DBUS[0..7] data bus, CLK, CE, IE, DOE, AOE, and RESET inputs. It has an ABUS[0..7] output and an 8-bit register with unknown values (indicated by question marks).
- Input Multiplexers:** Two multiplexers are shown. The first takes REALTIME/GRAPH and a control signal (0) to produce CLK, CE, IE, DOE, AOE, and RESET signals. The second takes REALTIME/GRAPH and a control signal (0) to produce MSB and LSB signals for the 8-bit register.
- Output Multiplexers:** Two multiplexers are shown. The first takes DBUS[0..7] and a control signal (0) to produce DBUS0 through DBUS7 signals. The second takes ABUS[0..7] and a control signal (0) to produce ABUS0 through ABUS7 signals.

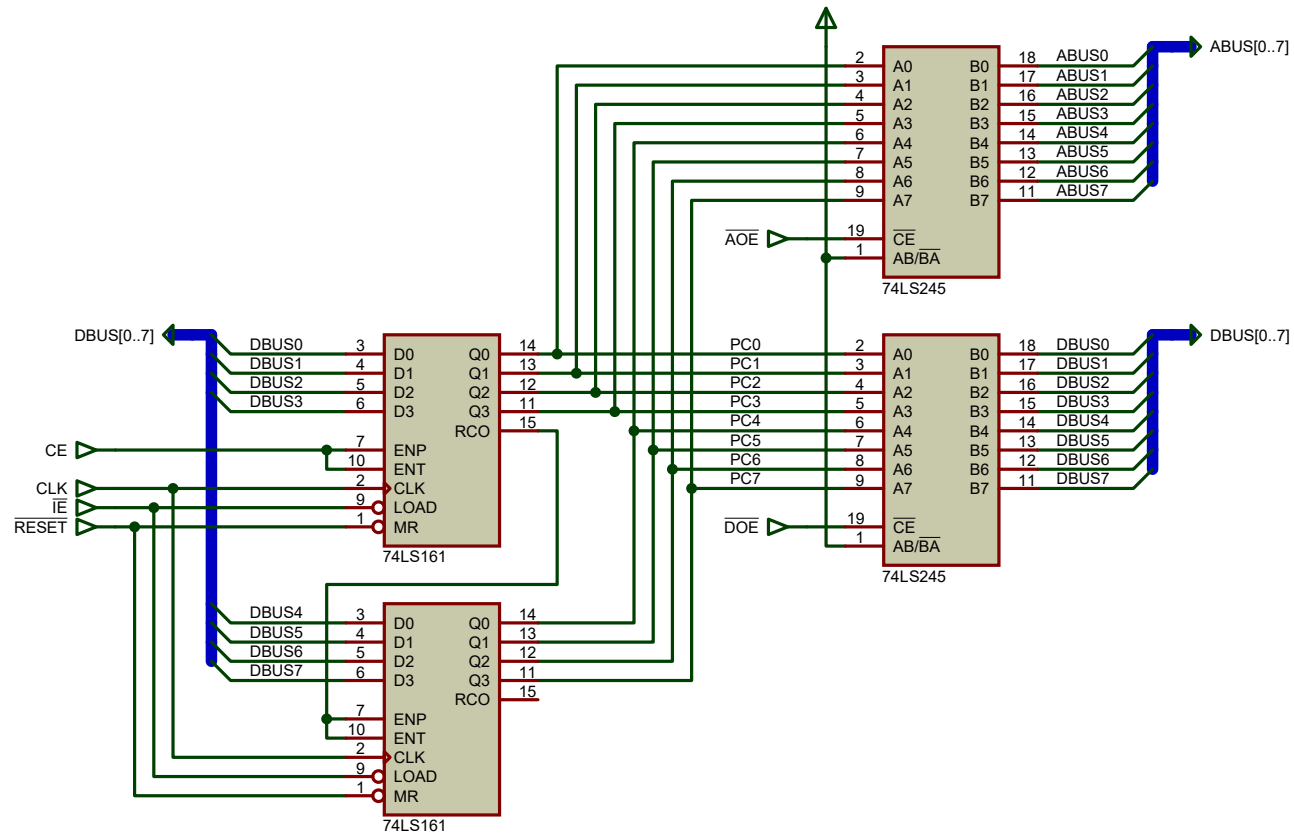
Timing Diagram:

The timing diagram shows the signals DBUS[0..7], ABUS[0..7], CLK, CE, IE, DOE, AOE, and RESET over a 32ms period. The signals are labeled as follows:

- DBUS[0..7]:** Data bus signals showing values 0E, FE, 02, 00, 00, 01, 02, 0E, 0F, 10, 11, 12, FE, FF, 00, 01, 02, 00.
- ABUS[0..7]:** Address bus signals showing values 00, 01, 02, 0E, 0F, 10, 11, 12, FE, FF, 00, 01, 02, 00.
- CLK:** Clock signal.
- CE:** Chip Enable signal.
- IE:** Input Enable signal.
- DOE:** Data Output Enable signal.
- AOE:** Address Output Enable signal.
- RESET:** Reset signal.

The timing diagram is labeled "Program Counter".

PROGRAM_COUNTER



TEST_PROGRAM_COUNTER

*SCRIPT GENERATOR TEST_PROGRAM_COUNTER

ALIAS ONOFF = \$ONS/OFF
 ALIAS IE = \$IES
 ALIAS DOE = \$DOES
 ALIAS AOE = \$AOES
 ALIAS RESET = \$RESET\$

PIN ONOFF
 PIN DBUS[0..7] = FLT
 PIN CE = FLT
 PIN IE = FLT
 PIN DOE = FLT
 PIN AOE = FLT
 PIN RESET = FLT

IF ONOFF = TRUE
 DBUS[0..7] = FLT
 CE = FLT
 IE = FLT
 DOE = FLT
 AOE = FLT
 RESET = FLT

ELSE
 CE = 0 AFTER 1m
 CE = 1 AFTER 5.5m
 IE = 1 AFTER 1m
 IE = 0 AFTER 8.5m
 IE = 1 AFTER 10.5m
 IE = 0 AFTER 18.5m
 IE = 1 AFTER 20.5m
 DOE = 1 AFTER 1m
 DOE = 0 AFTER 28.5m
 DOE = 1 AFTER 30.5m
 AOE = 1 AFTER 1m
 AOE = 0 AFTER 2.5m
 AOE = 1 AFTER 25.5m
 AOE = 0 AFTER 26.5m
 RESET = 0 AFTER 1m
 RESET = 1 AFTER 3.5m
 RESET = 0 AFTER 29.5m
 DBUS[0..7] = \$0E AFTER 7.5m
 SLEEP FOR 10.5m
 DBUS[0..7] = FLT
 DBUS[0..7] = \$FE AFTER 7m
 SLEEP FOR 10m
 DBUS[0..7] = FLT

ENDIF

END

*ENDSCRIPT

*SCRIPT GENERATOR TEST_PC_CLOCK

ALIAS ONOFF = \$ONS/OFF

PIN ONOFF
 PIN CLK = FLT

INT i

IF ONOFF = TRUE THEN
 CLK = FLT

ELSE

FOR i = 0 TO 15 STEP 1
 SLEEP FOR 1m
 CLK = 0
 SLEEP FOR 1m
 CLK = 1

NEXT i

ENDIF

END

*ENDSCRIPT

